Quadratic Equations

The big idea

A **quadratic equation** has the form **$ax^2 + bx + c = 0$**, where $a \neq 0$. Areas, projectile paths and many speed–time problems lead to quadratics. A quadratic has **at most two** roots.

ROOT OF A QUADRATIC

α is a root of $ax^2 + bx + c = 0$ if $a\alpha^2 + b\alpha + c = 0$. The roots of the equation are the zeroes of the polynomial $ax^2 + bx + c$.

Solving by factorisation (splitting the middle term)

Find two numbers whose **product is ac** and **sum is b**, then split bx and factor by grouping.

SOLVE $2X^2 - 5X + 3 = 0$

$ac = 6$, $b = -5$. The numbers -2 and -3 work (product 6, sum -5).

$2x^2 - 2x - 3x + 3 = 2x(x - 1) - 3(x - 1) = (2x - 3)(x - 1) = 0$.

So **$x = 3/2$ or $x = 1$**.

The quadratic formula

When factorising is hard, use the formula. It always works.

QUADRATIC FORMULA

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$$

SOLVE $X^2 + 4X - 5 = 0$ USING THE FORMULA

$a = 1, b = 4, c = -5$. $b^2 - 4ac = 16 + 20 = 36$.

$x = (-4 \pm 6)/2$, so **$x = 1$ or $x = -5$**.

Nature of roots: the discriminant

The expression **$D = b^2 - 4ac$** (the discriminant) tells you what kind of roots to expect without solving.

Value of D	Nature of roots
$D > 0$	Two distinct real roots
$D = 0$	Two equal real roots ($x = -b/2a$)
$D < 0$	No real roots

FIND K SO THAT $KX^2 - 6X + 1 = 0$ HAS EQUAL ROOTS

Equal roots $\Rightarrow D = 0$: $36 - 4k = 0 \Rightarrow$ **$k = 9$**.

Word problems

THE PRODUCT OF TWO CONSECUTIVE POSITIVE INTEGERS IS 306. FIND THEM.

Let them be x and $x + 1$. $x(x + 1) = 306 \Rightarrow x^2 + x - 306 = 0 \Rightarrow (x + 18)(x - 17) = 0$.

$x = 17$ (reject -18, it is not positive). The integers are **17 and 18**.

FORGETTING TO CHECK THE CONTEXT

A length, age or speed cannot be negative. Always reject roots that don't fit the situation, and say why.

Quick recap

- Standard form $ax^2 + bx + c = 0$, $a \neq 0$.
- Factorise by splitting the middle term, or use the quadratic formula.
- $D = b^2 - 4ac$ decides the nature of roots.

Practice questions

1. Which of these is a quadratic equation?
   (a) $x + 1/x = 2$ $(x \neq 0)$ written as it is (b) $(x + 1)^2 = 2(x - 3)$ (c) $x^3 - 4x^2 = 1$ (d) $(x + 2)^3 = x^3 - 4$

2. The roots of $x^2 - 3x - 10 = 0$ are
   (a) 5 and –2 (b) –5 and 2 (c) 5 and 2 (d) –5 and –2

3. The discriminant of $2x^2 - 4x + 3 = 0$ is
   (a) 8 (b) 40 (c) –8 (d) 16

4. For which value of k does $x^2 + kx + 9 = 0$ have equal roots?
   (a) 3 (b) 6 (or –6) (c) 9 (d) 0

5. If one root of $x^2 + 5x + k = 0$ is –2, then k is
   (a) 2 (b) 4 (c) 6 (d) –6

Answers: 1. b 2. a 3. c 4. b 5. c